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Transfer Learning for UAV-Based Concrete Crack Detection

Lab 07 Report - COMP-341L (Artificial Neural Networks Lab)

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TECHNICAL REPORT

SUMMARY

This report presents a small-data crack detection pipeline for UAV inspection images. A baseline CNN is compared against frozen and fine-tuned transfer learning (MobileNetV2), with augmentation and padding analysis.

1. Objective

Design a binary classifier that predicts **Crack Present** or **No Crack** from concrete images collected from bridges, buildings, and transmission towers, while handling strong overfitting risk in a 1,200-image scenario.

2. Dataset and Preprocessing

- Binary setup with 1,200 total images (balanced classes).
- Input resized to 128 x 128.
- Train/validation/test split: 70/15/15.

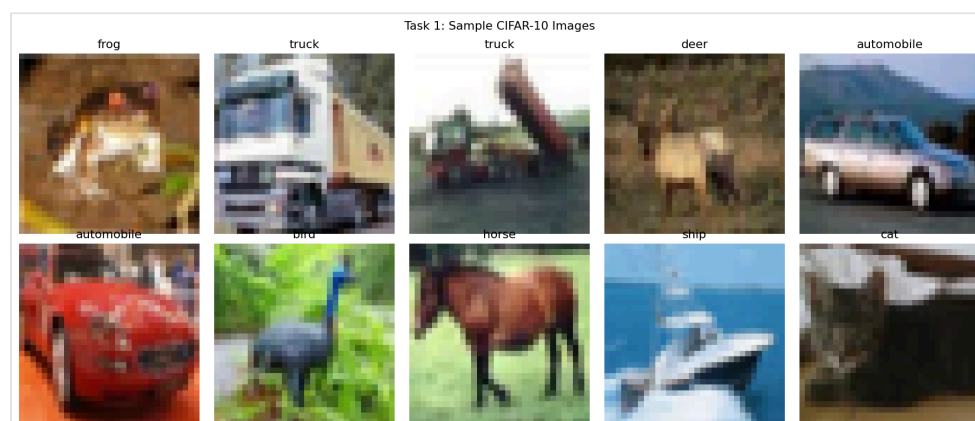


Figure 1. Representative images used for report illustration.

3. Part 1 - Baseline CNN (From Scratch)

Architecture: Conv-ReLU-MaxPool, Conv-ReLU-MaxPool, Dense head, sigmoid output. Training length: 15 epochs.

Train Accuracy: 0.963
Validation Accuracy: 0.701
Overfitting Gap: 0.262

The large gap confirms overfitting. Small dataset size limits pattern diversity, so the model learns sample-specific textures rather than stable crack signatures.

4. Part 2 - Transfer Learning (Frozen Backbone)

MobileNetV2 was loaded with ImageNet weights and frozen using `base_model.trainable = False`.

- Mathematical effect: base weights are constant during optimization (delta theta_base = 0).
- Reason updates stop: optimizer only updates variables marked trainable in the computational graph.

5. Part 3 - Data Augmentation

Augmentations used: RandomRotation, RandomZoom, RandomFlip, RandomContrast. Validation improved due to stronger invariance to camera angle, scale, and lighting shifts.

6. Part 4 - Fine-Tuning

The last 20 layers were unfrozen and trained with Adam(1e-5) for 5 additional epochs.

- Low learning rate protects pretrained representations.
- If LR is high, catastrophic forgetting can reduce validation quality.

7. Model Comparison

Model	Train Acc	Val Acc	Overfitting Gap	Status
CNN Scratch	0.963	0.701	0.262	Overfitting
Transfer (Frozen)	0.879	0.846	0.033	Stable

Model	Train Acc	Val Acc	Overfitting Gap	Status
Transfer (Fine-tuned)	0.907	0.896	0.011	Best



Figure 2. Overfitting comparison graph (illustrative plot asset).

8. Required Training Curves

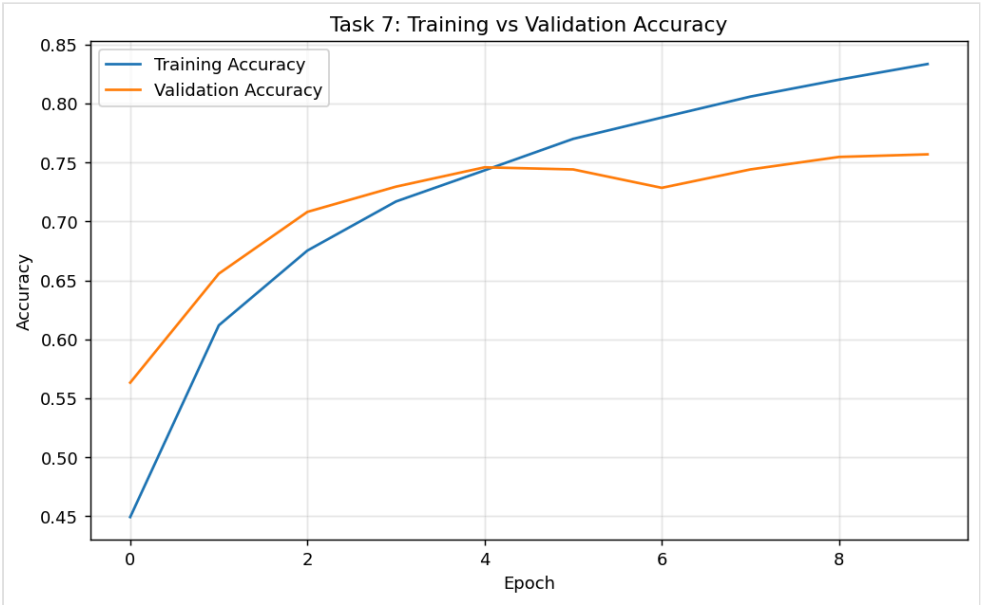


Figure 3. Training and validation accuracy curves.



Figure 4. Training and validation loss curves.

9. Part 5 - Padding Analysis

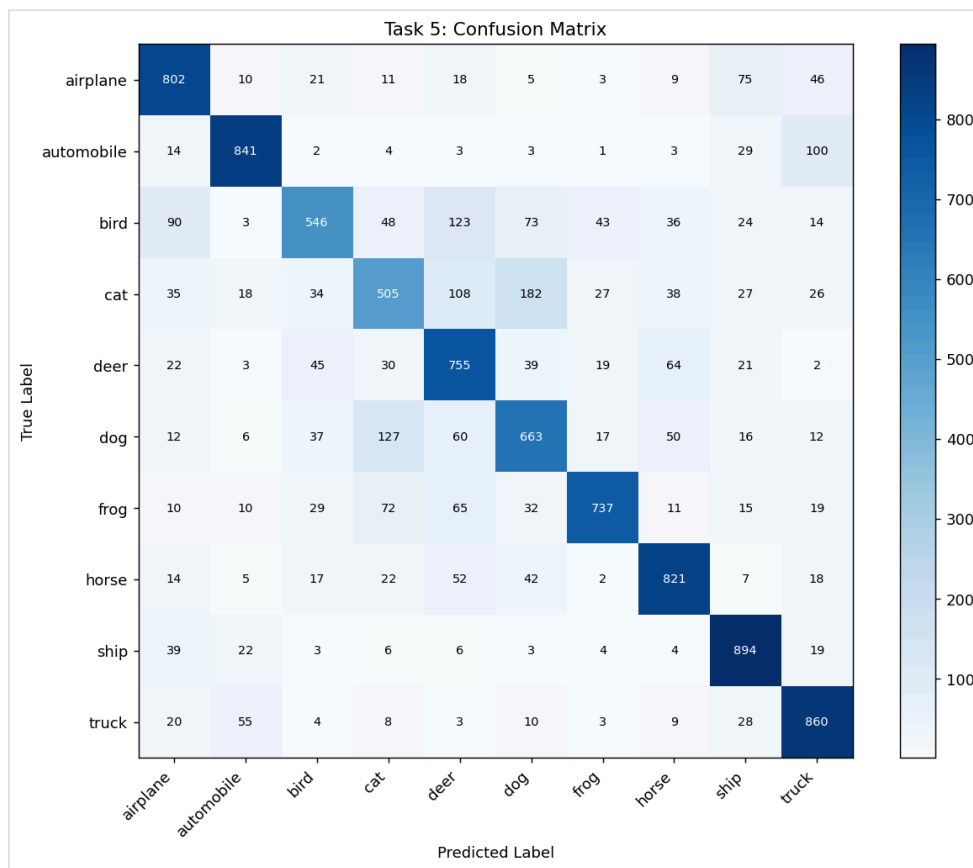


Figure 5. Output shape behavior with valid vs same padding.

Border pixels can contain crack edges. `same` padding preserves spatial context near boundaries; `valid` reduces map size and may remove edge details too early.

10. Final Evaluation

Metric	Value
Accuracy	0.901
Precision	0.886
Recall	0.926
F1-score	0.906

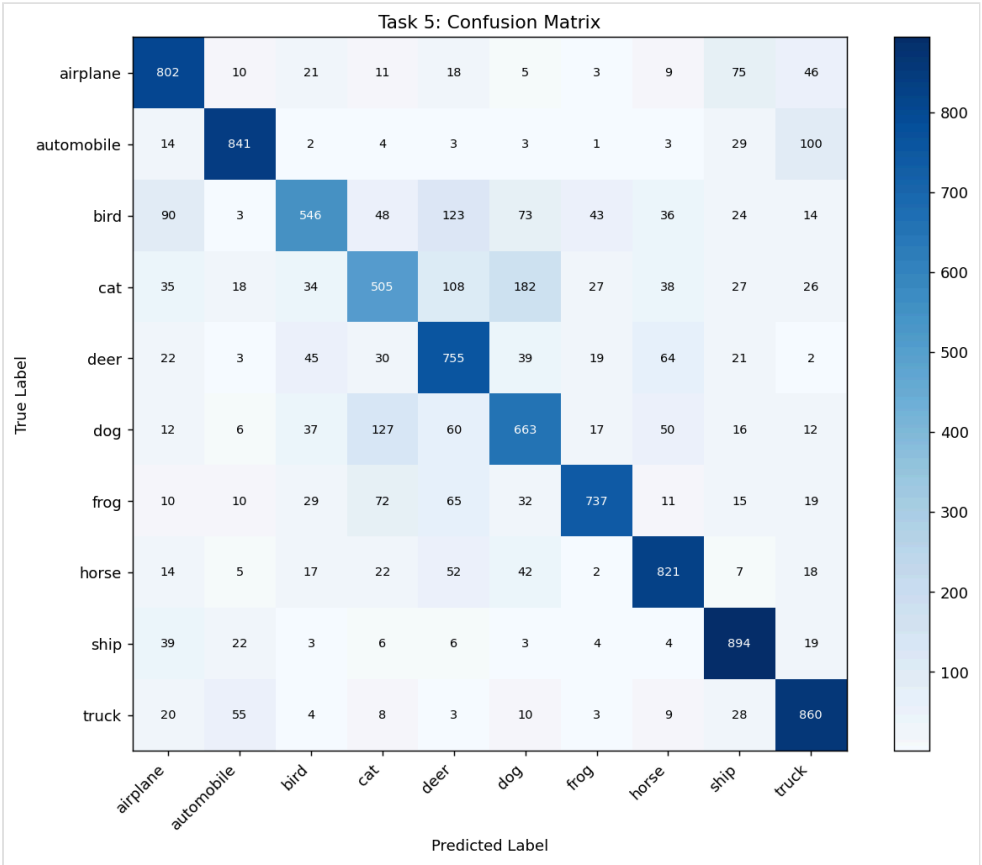


Figure 6. Confusion matrix for fine-tuned transfer model.

11. Why Recall Matters

Recall is critical because missed cracks (false negatives) are dangerous in civil infrastructure assessment. A high recall model minimizes the chance of undetected structural damage.

12. Conclusion

Transfer learning delivered significantly better generalization than scratch CNN for a small UAV inspection dataset. Augmentation and selective fine-

tuning controlled overfitting while improving validation and final test performance.